MERN Stack Developer

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- 3+ years of experience as a MERN Stack Developer, skilled in building scalable web applications with MongoDB, Express.js, React.js, and Node.js. Proficient in designing RESTful APIs, optimizing performance, and deploying with CI/CD pipelines and Docker. Experienced in Redis caching, Next.js, and React Query for state management.

CAREER OBJECTIVE

To leverage my 3+ years of experience as a MERN Stack Developer in building scalable web applications and designing efficient RESTful APIs. I aim to contribute to innovative projects, enhance user experiences, and continuously expand my technical expertise in modern web technologies.

PROFESSIONAL SYNOPSIS

- A detail-oriented professional with 3+ years of experience as a MERN Stack Developer, skilled in building scalable and high-performance web applications.
- Proficient in MongoDB, Express.js, React.js, and Node.js with expertise in designing and implementing RESTful APIs.
- Hands-on experience in optimizing database performance and deploying applications using CI/CD pipelines and Docker.
- Strong knowledge of Redis caching, Next.js for server-side rendering (SSR), and React Query for efficient state management.
- Adept at collaborating with cross-functional teams, mentoring developers, and enhancing user experiences.
- Experienced in deploying applications using Nginx and Apache servers with automated testing and deployment.
- Skilled in implementing dynamic routing, API routes, and managing bulk email queues using Bull MQ.
- Familiar with version control tools like Git and GitLab, and utilizing React Hook Form for efficient UI development

Personal Projects

1. ChainMind AI – Blockchain × AI Platform :

- Developed a full-stack platform combining **Blockchain and Generative AI** as core primitives.
- Designed and deployed custom **ERC-20 token (SmartAIToken)** for tokenized AI usage.
- Integrated OpenAI APIs to deliver AI services with **on-chain usage tracking**.
- Built smart contracts for AI service billing, access control, and transparency.
- Created scalable MERN architecture supporting **AI tools, smart contracts, and dashboards**.
- Delivered a production-ready platform showcasing real-world AI + blockchain utility.

2. Food Delivery App (MERN + React Native):

- Developed a full-stack platform combining **Blockchain and Generative AI** as core primitives.
- Designed and deployed custom **ERC-20 token (SmartAIToken)** for tokenized AI usage.
- Integrated OpenAI APIs to deliver AI services with **on-chain usage tracking**.
- Built smart contracts for AI service billing, access control, and transparency.
- Created scalable MERN architecture supporting **AI tools, smart contracts, and dashboards**.
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3. Personal Finance Management System

- Built a comprehensive personal finance application for **income, expense, and budget tracking**.
- Designed interactive dashboards with **financial analytics and visual reports**.
- Implemented secure authentication using **JWT and bcrypt**.
- Developed categorized transaction management and exportable reports (CSV/Excel).
- Enabled users to gain financial visibility, helping improve savings by **15–20%**.

4. Instagram Business Lead Automation Bot

- Built an automated lead generation system for identifying **Instagram business profiles**.
- Implemented industry filtering, location targeting, and data extraction pipelines.
- Exported leads to CSV/Excel for sales teams.
- Saved **20+ hours per week** through automation.

PROFESSIONAL EXPERIENCE

1. Working with Biz Technologies, Powai as Sr. Mern Stack Developer
(April 1st, 2025 to still working

Project Descritpion :-
At Biz Technologies Ltd., I contributed as a full-stack blockchain developer working on cutting-edge Web3-based platforms. My role involved developing decentralized applications, integrating smart contracts across frontend and backend layers, and modernizing legacy systems to leverage scalable, maintainable JavaScript architecture. The projects spanned a diverse set of domains including DeFi, NFT, social platforms, trading bots, and gamified reward systems.
Key projects included:

1. **Decentrawood – Web3 Community Platform**
    - Designed and developed a full-stack Web3 community platform with a **10-level referral system** and token-based incentives.
    - Integrated **Ethereum-compatible smart contracts** across BNB Chain and Polygon using **Ethers.js & Web3.js**.
    - Implemented **secure backend validation** for token holdings, referrals, and reward distribution.
    - Built scalable REST APIs using **Node.js, Express.js, and MongoDB** with optimized database schemas.
    - Automated CI/CD pipelines using **Docker and Jenkins**, ensuring reliable production deployments.
    - Delivered a production-ready system handling **1K+ daily blockchain transactions** with **99.9% uptime**.
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2. **Music Decentrawood – Web3 Music Platform**
    - Developed a decentralized music streaming platform enabling **direct artist-to-listener engagement** using blockchain.
    - Implemented **wallet-based authentication** and token-based reward distribution via smart contracts.
    - Built creator dashboards for **music analytics, engagement tracking, and royalty payouts**.
    - Integrated **Next.js + MERN backend** for scalable content delivery and metadata management.
    - Designed secure APIs for music uploads, playlists, and reward validation.
    - Enabled **85% revenue share for artists**, significantly higher than traditional platforms.

3. **Token-Based Game Wallet System (Polygon)**
   - Built a blockchain-based gaming wallet system with **daily login rewards and betting logic**.
   - Designed automated **token distribution logic** (platform fee, owner fee, burn mechanism).
   - Integrated smart contracts on **Polygon network** for secure token transfers and claims.
   - Implemented Redis caching to optimize wallet and transaction performance.
   - Developed backend APIs for **bet placement, result validation, and wallet updates**.
   - Managed real-world token economics with **2K+ active wallets**.

4. **DexBot – Automated DeFi Trading System**
   - Built an automated DeFi trading bot for **PancakeSwap (BSC)** with buy/sell execution logic.
   - Integrated **ParaSwap API** for optimal trade routing and price efficiency.
   - Developed real-time monitoring dashboards using **React and Socket.io**.
   - Implemented secure, non-custodial wallet interactions using **Ethers.js**.
   - Achieved **95% transaction success rate** with sub-second response times.

5. **Decentrawood Market – Web3 Prediction Market**

   - Designed and developed a **production-ready Polymarket-style prediction market backend** using **Node.js, Express.js, MongoDB, and Redis**.
   - Implemented **JWT-based authentication with refresh tokens**, wallet-based Web3 login, and role-based access control (user/admin).
   - Built complete **market lifecycle management** including market creation, real-time pricing, resolution, and automated payout calculation.
   - Developed **buy/sell trading engine** with AMM pricing logic, slippage protection, position tracking, and trade history.
   - Implemented **portfolio & P&L tracking** with realized/unrealized profit calculations and ROI analytics.
   - Integrated **Ethers.js (v6)** for wallet authentication, token approvals, and blockchain transaction validation.
   - Designed secure **payout and claim system** with expiration handling, transaction tracking, and status management.
   - Built **admin dashboards and analytics** for market control, user moderation, and system monitoring.
   - Secured APIs using **Zod validation, rate limiting, Helmet, bcrypt, and wallet signature verification**.
   - Containerized the system using **Docker & Docker Compose**, with Redis caching, background jobs (Bull), Swagger API docs, and Jest testing.

2. Working with CHANGE Networks, Ghansoli as a Jr. Mern Stack Developer (May 23rd, 2024 to March 30th, 2025)

Project : PHP to MERN Migration
Domain : Networking Device
Role : Jr. Mern Stack Developer

Project Description

CHANGE Networks is a leading networking solutions provider that specializes in supplying a wide range of networking devices, including routers, switches, firewalls, and other advanced networking products. The company focuses on delivering high-performance and secure networking solutions to businesses, ensuring seamless connectivity and robust infrastructure. With a commitment to innovation and customer satisfaction, CHANGE Networks supports enterprises in optimizing their network performance and enhancing overall operational efficiency.

Responsibilities

- Collaborating with stakeholders and team members to understand the existing PHP-based system and define the roadmap for migrating to the MERN stack.
- Analyzing and converting the backend logic from PHP to Node.js and Express.js while ensuring seamless functionality.
- Migrating the MySQL database to MongoDB and ensuring accurate data transfer and proper storage of previous records.
- Designing and implementing RESTful APIs for smooth communication between the frontend and backend.
- Developing dynamic and responsive UI components using React.js and enhancing the user experience with Tailwind CSS and MUI.
- Implementing caching mechanisms using Redis to optimize data retrieval and system performance.
- Integrating CI/CD pipelines for automated testing and deployment on Nginx and Apache servers.
- Conducting code reviews, identifying performance bottlenecks, and troubleshooting issues to ensure a smooth transition.
- Adding new features and functionalities to enhance the system's capabilities and meet user requirements.
- Maintaining proper documentation of the migration process and ensuring data integrity throughout the transition.
- Coordinating with cross-functional teams for seamless communication and timely project delivery.

3. MERN Stack Developer at Ganishka Enterprises.

Role : Jr. MERN Stack Developer

Responsibilities

- Conducted MERN stack training sessions, covering MongoDB, Express.js, React.js, and Node.js to enhance students' technical skills.
- Guided students in building full-stack web applications, ensuring best practices in coding and development.
- Designed and implemented RESTful APIs for seamless data communication between frontend and backend.
- Improved UI/UX using Tailwind CSS and MUI to create responsive and user-friendly interfaces.
- Assisted in deploying applications using Docker and CI/CD pipelines for automated builds and deployments.
- Monitored students' progress, provided feedback, and resolved technical challenges.
- Collaborated with team members to optimize project workflows and performance.
- Maintained documentation for project development and troubleshooting processes.

TECHNICAL COMPETENCIES

- **Frontend Technologies:** React.js, Next.js, Redux, React Query, React Hook Form, Tailwind CSS, Material UI (MUI), Bootstrap, TypeScript, Framer Motion.
- **Backend Technologies:** Node.js, Express.js, RESTful APIs, Bull MQ, Multer (File Upload & Processing), Authentication (JWT), Role-Based Access Control.
- **Databases:** MongoDB, MySQL, Redis, Data Modeling for High-Scale Systems, Query Optimization.
- **Blockchain & web3:** Web3.js, Ethers.js, Smart Contract Integration, Tokenomics Logic, Wallet Systems, BNB Chain & Polygon Networks, Trustless Verification, Token Distribution & Burns.
- **Version Control:** Git, GitLab, Github.
- **Cloud & Deployment Platforms:** Netlify, Render, Vercel.
- **Programming Languages:** JavaScript, TypeScript, Python, Java.
- **DevOps :** CI / CD Pipeline, AWS EC2, Docker, Jenkins, Nginx, Apache
- **Other Tools & Technologies:** Postman, CI/CD Pipeline, Redis Caching, Dynamic Routing, API Integration, Redis , Jira .

ACHIVEMENTS:

- Recognized as the Best NSS Volunteer for outstanding dedication to social service and community outreach programs.
- Awarded Best Camp Leader for managing a 7-day residential NSS camp, organizing events, and leading teams.
- Successfully organized and managed a 2-day college fest, ensuring smooth execution and team coordination.

ACADEMICS CREDENTIALS

- Perceiving Masters in IT from Mumbai University
- Completed BSC-IT from Mumbai University in 2024 with 9.17 CGPA.
- Completed HSC from Mumbai University in year 2021 with 80%.
- Completed SSC from Maharashtra board in year 2019 with 71.40%.

PERSONAL INFORMATION

➢ **Date of Birth** : 24th January, 2003

➢ **Language** : Hindi, English, Marathi

➢ **Permanent Address** : PMGP Colony, Mankhurd (w), Mumbai-43

I hereby declare that all the above information given by me is true of my best knowledge.

Date:
Place: Mankhurd, Mumbai (Vaibhav Vasant Patil)